

OPEN BY DESIGN Open Science

IS, DS, AI, and SD

A concise view of principles, infrastructures, social responsibility, and Brazil's strategic agenda.

Washington Segundo

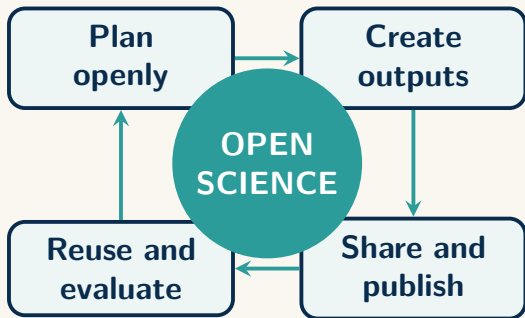
Coordenador-geral de Informação Científica e Técnica

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Open science redesigns the entire research cycle

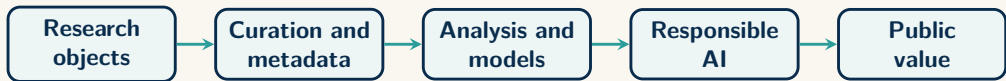


A PARADIGM SHIFT

- Openness begins at planning, not after publication.
- Publications, data, code, methods, and reviews become reusable outputs.
- Governance, preservation, and participation are part of scientific quality.

Core idea: knowledge should be accessible, auditable, interoperable, and useful to society.

Information Science is the foundation of reliable AI



Data, publications, code Context and provenance Patterns and evidence Grounded automation Decisions and services

Information Science

Organizes, curates, represents, and
mediates knowledge

Data Science

Analyzes patterns, models
phenomena, and extracts value

Artificial Intelligence

Automates cognition, prediction, and
language interaction



A long evolution led from private exchange to open ecosystems



Digital networks separated the traditional functions of journals: record, certification, dissemination, and preservation.



Ibict has shaped scientific information since 1954

1954

Founded as Brazil's national
institute dedicated to
bibliography and scientific
documentation

Today

A research unit of the
Ministry of Science,
Technology and Innovation
- MCTI

A LONG-STANDING PUBLIC MISSION

- Organize, preserve, and disseminate Brazil's scientific and technological information.
- Build national services, standards, and interoperable information infrastructures.
- Connect Brazilian research with regional and international knowledge networks.

A STRONG TRADITION OF OPENNESS Ibict has played a sustained leadership role in **Open Access** and **Open Science**, advancing repositories, aggregators, research data, digital preservation, persistent identifiers, and public-interest governance.



The ecosystem is broader than open access

Open outputs

Publications, research data, software, methods, and educational resources.

Open processes

Preprints, transparent peer review, responsible assessment, and reproducible workflows.

Open infrastructures

Repositories, identifiers, metadata standards, preservation, and research information systems.

Open participation

Citizen science, knowledge dialogue, inclusion, and public engagement.

Infrastructure connects every pillar and keeps openness sustainable.

Four principles balance technical value and social responsibility

FAIR

Make data findable, accessible, interoperable, and reusable for people and machines.

CARE

Protect collective benefit, authority, responsibility, and ethics in data governance.

TRUST

Build transparent, responsible, user-focused, sustainable, and technically sound repositories.

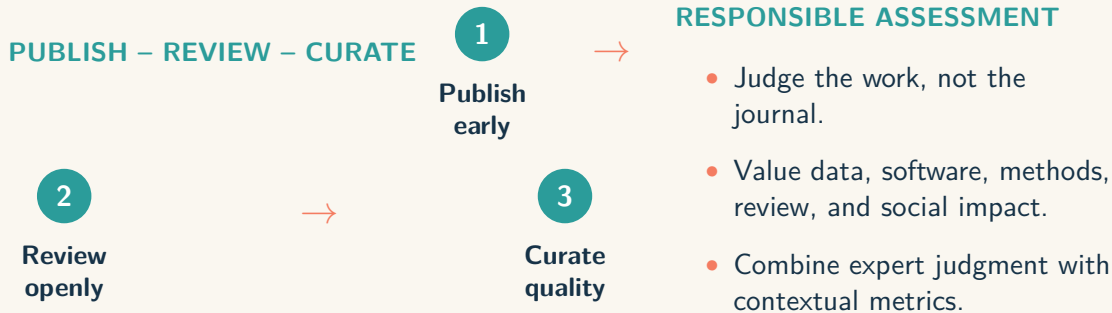
DEIA

Embed diversity, equity, inclusion, and accessibility in research systems.

Rule of thumb: as open as possible, as closed as necessary.



Communication and assessment are moving toward transparency



Preprints accelerate circulation; open reviews expose reasoning; curation adds context and certification.

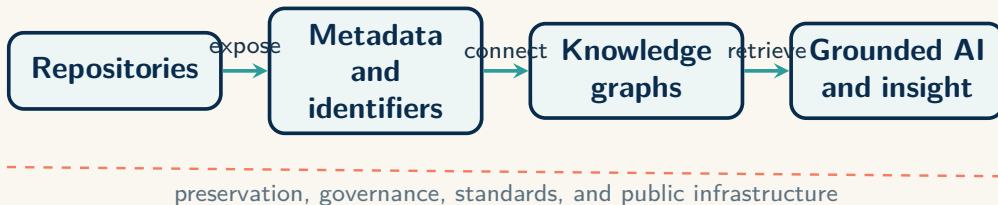
Open access models distribute cost and power

Route	How it opens access	Main strength	Main risk
Green	Deposit in a repository	Uses public infrastructure and can avoid publication fees	Embargoes or restrictive publisher policies
Gold	Immediate access on the publisher platform	The version of record is openly available	APCs can shift exclusion from readers to authors
Diamond	No fees for readers or authors	Aligns publishing with the public interest	Requires durable institutional funding and shared governance

Publicly funded research creates maximum public value when access, publishing, and governance remain equitable.



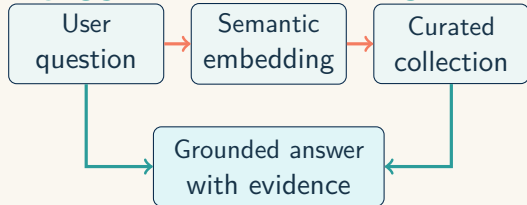
Trustworthy AI begins with trustworthy research infrastructure



Preservation and interoperability turn stored objects into reusable evidence.

Semantic systems connect meaning, not only keywords

A GROUNDED RETRIEVAL FLOW



WHAT MAKES IT WORK

- Controlled vocabularies and ontologies reduce ambiguity.
- Multilingual embeddings retrieve concepts across languages.
- RAG grounds generated answers in real documents.
- Rich metadata makes sources traceable and reusable.

Openness must also deliver informational justice

THREE DIMENSIONS

- **Distribution:** equitable access to information and infrastructure.
- **Recognition:** respect for diverse people, knowledge, and epistemologies.
- **Participation:** meaningful inclusion in producing and using knowledge.

ETHICAL SAFEGUARDS

- Consent and risk assessment
- Privacy and controlled access
- Collective and Indigenous data rights
- Protection from harmful or dual use



Libraries are becoming critical knowledge infrastructure

AN INSTITUTIONAL EVOLUTION

1 Collections

Acquire and provide access

2 Digital services

Connect repositories and systems

3 Infrastructure

Steward data and long-term access

THE PROFESSIONAL ROLE EXPANDS

- Data stewardship and metadata enrichment
- Digital preservation and authenticity
- Licensing, privacy, and ethical guidance
- Persistent identifiers and interoperability
- Digital literacy and public mediation
- Coordination across research, IT, government, and society



Implementation depends on human mediation

INCLUSIVE DESIGN Build for diverse communities

- Accessible interfaces
- Multilingual discovery
- Disciplinary flexibility
- Context-sensitive workflows

GOVERNANCE Define responsibility

- Data control and privacy
- Curation criteria
- Ethical safeguards
- Sustainable ownership

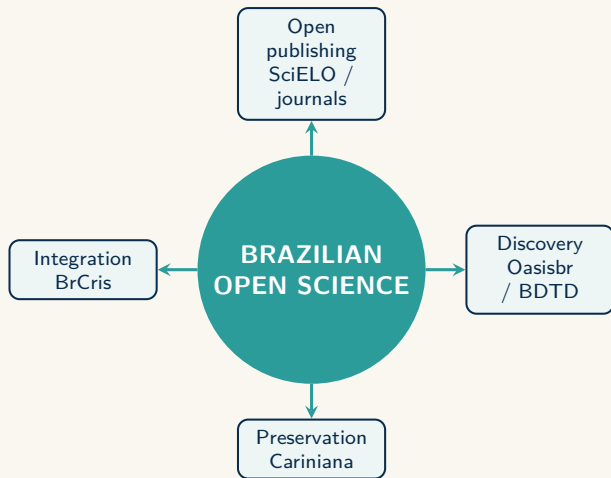
DIGITAL LITERACY Enable critical use

- Practical training
- Limits and uncertainty
- Reproducible practices
- Responsible AI use

Multidisciplinary support teams translate Open Science principles into everyday research practice.



Brazil already has the foundations of a national ecosystem



STRATEGIC ASSETS

- SciELO and open journals
- BDTD and Oasisbr
- BrCris and research integration
- Cariniana and digital preservation
- Data, citizen science, and software networks

The next step is coordinated governance, stable funding, and stronger interoperability.

Ibict connects the national ecosystem in layers

COLLECT AND PUBLISH

- Institutional repositories and digital libraries
- Brazilian Digital Library of Theses and Dissertations
- Journal and publication support services

AGGREGATE AND CONNECT

- Oasisbr for unified discovery
- National repository networks and shared standards

ANALYZE AND GOVERN

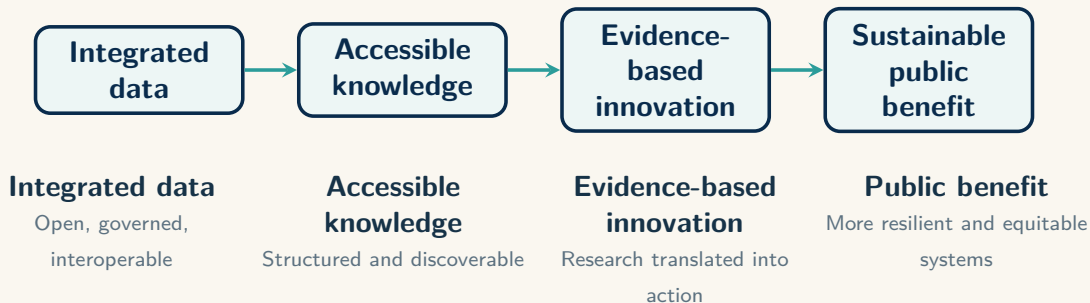
- BrCris for research information and policy intelligence
- Laguna for semantic integration and large-scale analysis

PRESERVE AND IDENTIFY

- Cariniana for distributed digital preservation
- dARK for persistent, traceable identifiers

Together, these layers preserve institutional autonomy while enabling national-scale

Scientific infrastructure enables sustainable development



Three complementary infrastructures strengthen Brazilian research

BRCRIS Connects the research system

- Researchers and institutions
- Projects and funding
- Outputs and infrastructures

Enables visibility, governance, assessment, and evidence-based policy.

REDE DARK Keeps resources identifiable

- Persistent ARK identifiers
- Traceability and citation
- Long-term access

Protects reproducibility and institutional memory from broken links.

PROJECT LAGUNA Integrates heterogeneous data

- Semantic enrichment
- Interoperability
- Large-scale discovery

Transforms isolated sources into a shared analytical environment.

BrCris creates a national view of Brazilian research

BRAZILIAN CURRENT RESEARCH INFORMATION SYSTEM

- Developed and led by **Ibict**; officially established in 2014.
- Integrates, retrieves, certifies, and visualizes information about Brazil's research ecosystem.
- Organizes fragmented records into a unified national model aligned with international standards.
- Provides free access for researchers, managers, funders, journalists, and citizens.

15.9M

publications indexed

Platform count displayed in July 2026

A public gateway to understand who produces knowledge, where, with whom, and with what results.

Source: brcris.ibict.br and official BrCris About page, accessed 28 Aug. 2026.



BrCris connects actors, outputs, and infrastructures

RESEARCH ACTORS

- Researchers
- Institutions
- Research groups
- Graduate programs
- Funders and projects

RESEARCH OUTPUTS

- Publications
- Theses and dissertations
- Scientific datasets
- Patents
- Software

INTEGRATED SOURCES

- Lattes and CNPq directories
- Oasisbr and Capes/Sucupira
- OpenAlex and OpenAIRE
- DOAJ, INPI, and Espacenet
- NDLTLD, ROR, and Wikidata

Source: official BrCris About page and FAQ, accessed 28 Aug. 2026.



Integration turns records into policy intelligence

INFORMATION PIPELINE

1. Collect records from national and international sources.
2. Standardize, normalize, and deduplicate the metadata.
3. Connect people, organizations, projects, and results.
4. Provide search, individual profiles, and indicator dashboards.

DECISIONS IT CAN SUPPORT

- Map collaboration networks and scientific communities.
- Compare institutional and thematic research profiles.
- Identify strengths, gaps, and emerging trends.
- Improve transparency, visibility, and research assessment.
- Ground science, technology, and innovation policies in evidence.

Source: official BrCris About page, Dashboards, and FAQ, accessed 28 Aug. 2026.

Oasisbr is Brazil's open research gateway

ONE SEARCH ACROSS OPEN SOURCES

- A multidisciplinary discovery service developed by Ibict.
- Aggregates open content from journals, publication repositories, research data repositories, and digital libraries.
- Provides free access through one interface while directing users to the full object at the source institution.
- Exchanges content with Portuguese infrastructure through RCAAP.

CONTENT TYPES

- Journal articles and preprints
- Books and book chapters
- Conference papers
- Research datasets
- Theses and dissertations
- Undergraduate works and reports

Source: Oasisbr official portal, About, and FAQ, accessed 28 Aug. 2026.

BDTD gives visibility to graduate research

A NATIONAL SERVICE SINCE 2002

- Conceived, developed, and maintained by Ibict.
- Integrates theses and dissertations from Brazilian teaching and research institutions.
- Also includes records of works defended by Brazilians at institutions abroad.
- Offers free national discovery; full texts remain hosted by participating institutions.

PUBLIC VALUE

- Visibility for postgraduate research
- Shared metadata standards
- Institutional interoperability
- Access to specialized knowledge
- Governance of public investment in graduate education

Source: BDTD official About page, FAQ, and portal, accessed 28 Aug. 2026.



Oasisbr, BDTD, and BrCris play distinct roles

Service	Primary question	Scope	Strategic role
Oasisbr	What open research outputs can I discover?	Publications, data, books, conference works, theses, and more	Multidisciplinary access and national/international aggregation
BDTD	What graduate research has been produced?	Brazilian theses and dissertations, including records from abroad	Visibility and governance of postgraduate knowledge
BrCris	How is the research ecosystem connected?	Actors, organizations, projects, programs, outputs, patents, and software	Research intelligence, indicators, governance, and policy support

Together they connect **discovery**, **specialized graduate knowledge**, and **system-level intelligence**.

Sources: official Oasisbr, BDTD, and BrCris portals, accessed 28 Aug. 2026.

dARK turns persistence into regional sovereignty

WHAT IT ADDS TO ARK

- **Decentralization:** participating institutions share operational responsibility.
- **Traceability:** identifier events remain auditable over time.
- **Resilience:** distributed governance reduces single points of failure.
- **Sovereignty:** regional institutions define policies and priorities.

HOW IT WORKS WITH OASISBR

1. Repositories expose qualified metadata through open protocols.
2. Oasisbr aggregates and validates the records.
3. dARK assigns persistent identifiers to eligible objects.
4. Users discover content centrally while access remains with the source institution.



Open infrastructures advance a focused set of SDGs

SDG 4 – Quality Education

Open scientific knowledge strengthens higher education, research, and information literacy.

SDG 9 – Innovation and Infrastructure

Robust, interoperable digital systems provide the backbone for scientific innovation.

SDG 10 – Reduced Inequalities

Open access reduces institutional, regional, and economic asymmetries in knowledge.

SDG 16 – Strong Institutions

Transparent information systems support accountability, ethical governance, and evidence-based decisions.

SDG 17 – Partnerships

Shared standards and interoperable networks enable national and international co-operation.

Cross-cutting effect: preservation and responsible reuse also support sustainable cities, institutions, and consumption of digital resources.



Applied AI creates value when the information base is ready

Application	AI capability	Public-interest value
ServicesPI	Semantic search + NLP	Natural-language discovery across complex patent collections
Patente.IA	Text + image similarity	Broader detection of related inventions and prior art
RDAPP	NLP + semantic retrieval	Structured access and analysis across public-policy documents
SAVITS	BI + ML + LLM	Monitoring, prediction, and explainable impact assessment

The recurring architecture is simple: **infrastructure** → **structured data** → **AI application**.

Brazil needs a national implementation plan

POLICY ARCHITECTURE

- A national policy with clear public-interest goals
- Defined roles for MCTI, funders, institutions, and scientific societies
- Shared standards with room for regional and disciplinary adaptation
- Stable funding for open infrastructures and professional teams

DELIVERY MECHANISMS

- Phased mandates linked to practical support
- Incentives aligned with responsible assessment
- Training, monitoring, and transparent indicators
- Continuous consultation with researchers and society

The objective is gradual, consistent change that increases the public return on research investment.

The agenda now shifts from isolated projects to durable policy

SYSTEM PRIORITIES

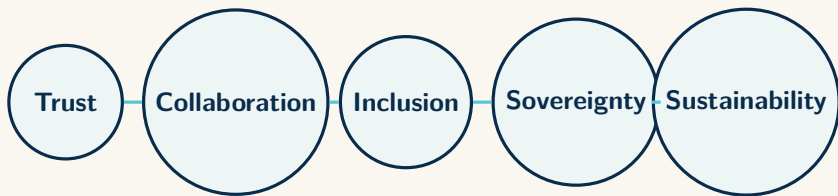
- Sustain public-interest infrastructures.
- Integrate platforms, standards, and preservation.
- Train researchers, librarians, data stewards, and developers.
- Reform incentives and research assessment.
- Govern quality, privacy, bias, and accountability.

RESEARCHER ACTIONS

- Publish through open routes.
- Share documented data and code.
- Use persistent identifiers and open licenses.
- Join transparent review and citizen science.
- Plan semantics, governance, and preservation from the start.



Open science is a strategic investment



Infrastructure + public policy + scientific practice + social movement

Together, they make research more reliable, reusable, inclusive, and aligned with the public interest.

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Obrigado!

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Guarani

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Quechua

Na gode

Hausa

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धन्यवाद

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நன்றி

Tamil

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Amharic

Terima
kasih

Indonesian

谢谢

Chinese

Thank you

English

